MICROCONTROLLERS

(PBCST504)

WATER LEVEL DETECTION USING STEM32

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ABSTRACT

The Water Level Overflow Indicator is designed to monitor the water level in a tank and provide a warning when the water approaches a predefined high level. The system uses an HC-SR04 ultrasonic sensor to measure the distance between the sensor and the water surface. An STM32 Blue Pill microcontroller processes the sensor readings and controls a 16×2 LCD, red and green LEDs, and a buzzer to indicate the water-level status. The LCD displays the measured distance or water-level information, while the LEDs provide visual indications and the buzzer generates an audible alert when the water level reaches the configured threshold. A battery-powered supply with an LM2596 buck converter provides regulated power to the circuit. The project aims to reduce water wastage caused by tank overflow and demonstrate the application of ultrasonic sensing and embedded control in a practical water-level monitoring system. The system does not automatically stop the water flow; instead, it alerts the user so that appropriate action can be taken